# Formal Laguages - 2

In the previous chapter we have defined $\Sigma^*$ and connected the terms

Question — Problem — Language. → a subset of $\Sigma^*$

We further showed that the set of 'all questions' is much much smaller than the set of 'all problems'.

In this chapter we shall delve deeper into Languages.

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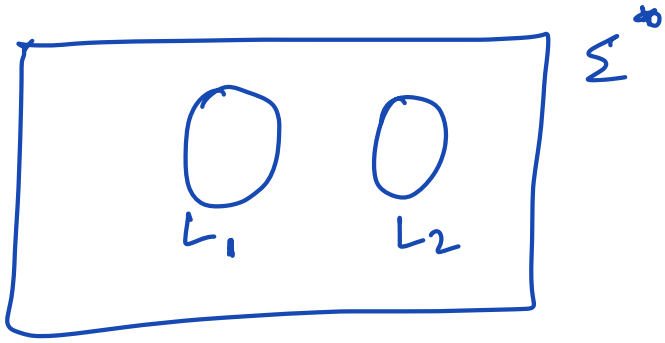
We define certain operations over languages.



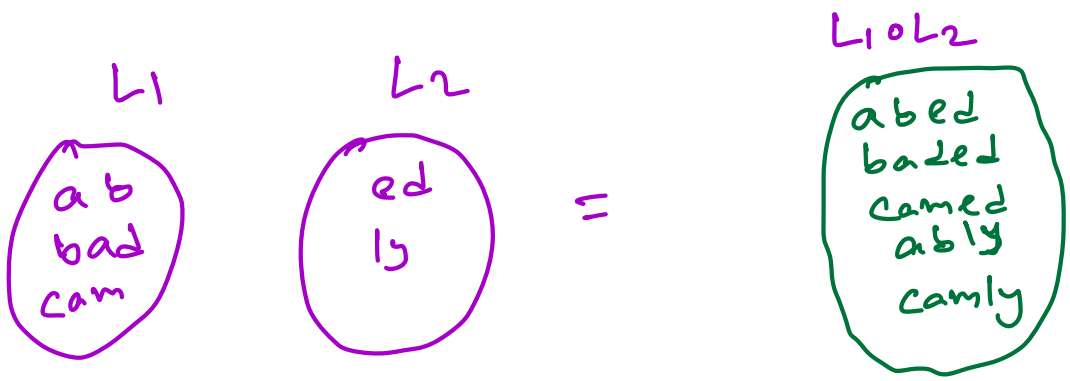

| $L_1 \cup L_2$ | $L_1 \cap L_2$ | $L_1^c$ |
|---|---|---|
| union | intersection | complementation |

follow from set theory.

In addition there are a few more operations

**Concatenation of languages**

For two languages $L_1$ & $L_2$, their concatenation $L_1 \circ L_2$ is defines as the set of all possible concatenations $x.y$ where $x \in L_1$ & $y \in L_2$.




Concatenation is similarly defined over infinite languages as well.

**n-th concatenation of a language**

$$L^n = L \circ L \circ L \cdots \circ L$$

For example, $L = \{ab, xy\}$

$$L^3 = \begin{matrix} ab \\ xy \end{matrix} \circ \begin{matrix} ab \\ xy \end{matrix} \circ \begin{matrix} ab \\ xy \end{matrix}$$

$$= \{abab,\ abxy,\ xyab,\ xyxy\} \circ \{ab,\ xy\}$$

$$= \{abab\,ab,\ ab\,xy\,ab,\ xyab\,ab,\ xyxy\,ab,\ abab\,xy,\ abxy\,xy,\ xyab\,xy,\ xyxy\,xy\}$$

## Kleene – Star operator :

Let $L$ be any language.

The Kleene star

$$L^* = L^0 \cup L^1 \cup L^2 \cdots$$

$$= \bigcup_{n \in \mathbb{N}} L^n$$

The union of all possible $n$-concatenations of $L$.

As a matter of fact, every time we write $\Sigma^*$, we are referring to the Kleene-star of $\Sigma$.

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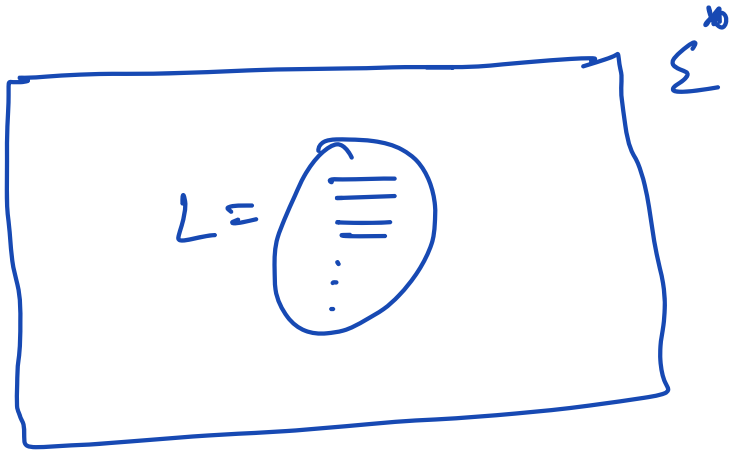


Among all possible languages over $\Sigma$, it would be too ambitious to assume one can solve the membership query

$$x \overset{?}{\in} L$$

for any arbitrary $L$.

As the first step, we shall only focus on languages that we can $\boxed{\text{build}}$.

But can we build a language?

By following a grammar.

# Grammar

Loosely speaking grammar is a bunch of rules by which a language is generated.

Imagine you are speaking to someone and you are in the middle of a sentence. If notice it consciously, what you are doing is to generate a sentence in a language while obeying a strict rules of grammar. This 'grammar' includes vocabulary as well as syntax rules.

Language generated by a grammar is therefore the set of all possible grammatically correct sentences (strings) generated within that grammar.

Now, we extend this notion of grammar to formal languages.

We define grammar precisely as a 4-tuple

$$G = (\Sigma, V, S, P)$$

where,

$\Sigma$: **Terminal alphabet**: a finite, non-empty set of symbols called terminals.

$V$: **Variables/Non-terminals**: a finite, non-empty set of symbols disjoint from $\Sigma$.

$S \in V$: The start variable

$P$: a finite set of productions (or rules) of the form $\alpha \longrightarrow \beta$.

where,

$$\alpha \in (\Sigma \cup V)^* V (\Sigma \cup V)^*$$

$$\beta \in (\Sigma \cup V)^*$$

which means,

$\alpha$ is a string over terminal alphabet and variables. $\alpha$ contains at least one variable.

$\beta$ is simply a string over terminal alphabet and variables. $\beta$ may not contain variables.

**Note:** A production rule $\alpha \to \beta$ should be read as – "alpha may be replaced with $\beta$".

Q. Ok, now what is the idea?

A. Given a grammar $G$, it produces strings and set of all these strings is the language generated by $G$.

start variable

**Example 1:** $G_1 = (\{0,1\}, \{S, T, 0, I\}, S, P)$

where $P =$

$S \to 0T$
$S \to 0I$
$T \to SI$
$0 \to 0$
$I \to 1$

what language does $G$ produce?

Ans:

① Start with the start variable.

$$S$$

② Look up the rules. S can be replaced with 0T or 0I. Make a choice.

$$S \downarrow 0T$$

③ Look up the rules. 0 can be replaced with 0 and T can be replaced with SI and there is no choice.

$$S \downarrow 0T \downarrow 0SI$$

④ This time we choose $S \to 0I$ and $I \to 1$

$$S \downarrow 0T \downarrow 0SI \to 0\,0I1 \to 0011$$

$\therefore\ 0011 \in L(G_1)$.

With some inspection, we can convince ourselves that $L(G_1) = \{0^n 1^n \mid n \geq 1\}$

<u>Example 2:</u> $G_2 = (\{0,1,2\}, \{S, A\}, S, P)$

with $P =$

$$
\begin{aligned}
S &\to 0SA2 \\
S &\to \epsilon \\
2A &\to A2 \\
0A &\to 01 \\
1A &\to 11
\end{aligned}
$$

what language does $G_2$ produce?

<u>Ans:</u>

$$
\begin{aligned}
&S && (S \to 0SA2) \\
&\downarrow \\
&0SA2 && (S \to 0SA2) \\
&\downarrow \\
&00SA2A2 && (S \to \epsilon,\ 2A \to A2) \\
&\downarrow \\
&00AA22 && (0A \to 01) \\
&\downarrow \\
&001A22 && (1A \to 11) \\
&\downarrow \\
&00 1122
\end{aligned}
$$

in fact $L(G_2) = \{0^m 1^n 2^n \mid m \geq 0\}$.

<u>Example:</u> $G_3 = (\Sigma, V, S, P)$

with $\Sigma = \{0, 1, 2, \ldots 8, 9, e, +, -, .\}$

$V = \{S, M, N, D\}$

$P =$ $S \rightarrow M.N \mid MeM \mid M.NeM.$

$M \rightarrow N \mid +N \mid -N$

$N \rightarrow DN \mid D$

$D \rightarrow 0 \mid 1 \mid 2 \ldots \mid 9$

what language does $G_3$ produce?

<u>Ans:</u>

$S \rightarrow M.NeM \rightarrow N.NeM \rightarrow DN.NeM$

$\rightarrow 1N.NeM \rightarrow 1D.NeM \rightarrow 12.NeM$

$\rightarrow 12.DeM \rightarrow 12.3eM \rightarrow 12.3e-N$

$\rightarrow 12.3e-4$

In fact $G_3$ produces scientific representation of nums.

**Example:** $G_6 = \{\{(,)\}, \{S\}, S, P\}$

$P =$
$$S \to (S)$$
$$S \to SS$$
$$S \to \epsilon$$

**Example:** $G_5 = (\Sigma, V, S, P)$

with $\Sigma = \{a, +, \times, (, )\}$

$V = \{S, T, F\}$ for $a \in \mathbb{N}$

$P$:
$$S \to T+F \mid T$$
$$T \to T \times F \mid F$$
$$F \to (S) \mid a$$

$L(G_5) = ?$

**Ans:** $S \to T \to T \times F \to F \times F$

$\to (S) \times (S) \to (T+F) \times (T+F)$

$\to (a+a) \times (a+a)$

In fact $G_5$ generates simple arithmetic formulas.